

Earth Science: Minerals and Rocks Study Guide

6th Grade Science

1 Fundamental Concepts

1.1 What is a Mineral?

A **mineral** is a naturally occurring, inorganic solid that has:

- A crystal structure
- A definite chemical composition
- Formed through geological processes

1.2 What is a Rock?

A **rock** is a naturally formed solid made of one or more minerals that make up the Earth's crust.

1.3 Key Differences

Minerals	Rocks
Single, pure substance	Mixture of minerals
Definite chemical formula	Variable composition
Crystalline structure	May or may not show crystals
Examples: quartz, calcite, gold	Examples: granite, sandstone, marble

2 Mineral Identification Properties

2.1 Physical Properties for Identification

1. Color

- Quick visual clue but often unreliable alone
- Same mineral can have different colors
- Different minerals can have the same color

2. Streak

- Color of mineral's powder when rubbed on streak plate
- More reliable than surface color
- Example: Pyrite has golden color but black streak

3. Luster

- How mineral reflects light
- Types: metallic, glassy, pearly, dull, waxy
- Example: Gold has metallic luster, quartz has glassy luster

4. Hardness

- Measure of how easily mineral can be scratched
- Tested using Mohs Hardness Scale (1-10)
- 1 = softest (talc), 10 = hardest (diamond)

5. Cleavage vs. Fracture

- **Cleavage:** Breaks along flat, planar surfaces
- **Fracture:** Breaks with irregular, uneven surfaces
- Example: Mica has perfect cleavage, quartz has conchoidal fracture

6. Density

- Mass per unit volume
- "Heft test" - how heavy it feels for its size
- Example: Gold feels much heavier than pyrite of same size

2.2 Mohs Hardness Scale

Scale	Mineral (Symbol)	Test
1	Talc	Can be scratched by fingernail
2	Gypsum	Can be scratched by fingernail
3	Calcite (CaCO_3)	Can be scratched by copper penny
4	Fluorite (CaF_2)	Can be scratched by iron nail
5	Apatite	Can be scratched by glass
6	Feldspar	Can scratch glass
7	Quartz (SiO_2)	Can scratch glass easily
8	Topaz	Can scratch quartz
9	Corundum (Al_2O_3)	Can scratch topaz
10	Diamond (C)	Can scratch everything

3 Special Tests for Mineral Identification

3.1 Additional Identification Methods

- **Acid Test:** Calcite fizzes when hydrochloric acid is applied
- **Magnetism:** Magnetite is attracted to magnets
- **Taste:** Halite (rock salt) tastes salty
- **Smell:** Sulfur has a distinctive smell

3.2 Common Mineral Examples

Mineral	Key Properties	Uses
Quartz (SiO_2)	Glassy luster, hardness 7, white streak	Glass, electronics
Calcite (CaCO_3)	Reacts with acid, hardness 3	Construction, antacids
Gold (Au)	Metallic luster, soft, golden streak	Jewelry, electronics
Pyrite (FeS_2)	Metallic luster, hard, black streak	Fool's gold, sulfur source
Mica	Perfect cleavage, flexible sheets	Electronics, insulation

4 Weathering and Erosion

4.1 Key Processes

1. Weathering

- Chemical and physical breakdown of rock at Earth's surface
- Creates sediment (small pieces of rock)

2. Erosion

- Movement of weathered material by water, wind, or ice
- Transports sediment to new locations

3. Deposition

- Laying down of sediment in new locations
- Forms layers that may become sedimentary rock

5 The Three Types of Rocks

5.1 Igneous Rocks

- **Formation:** From cooling of molten rock (magma or lava)
- **Location:** Magma cools underground, lava cools on surface
- **Examples:** Granite, basalt, obsidian
- **Characteristics:** May have crystals, no fossils

5.2 Sedimentary Rocks

- **Formation:** From compaction and cementation of sediments
- **Process:** Weathering → Erosion → Deposition → Compaction → Cementation
- **Examples:** Sandstone, limestone, shale
- **Characteristics:** May contain fossils, layered structure

5.3 Metamorphic Rocks

- **Formation:** From heat and pressure on existing rocks
- **Process:** Changes mineral structure without melting
- **Examples:** Marble, slate, gneiss
- **Characteristics:** May have foliation (banding), no fossils

6 The Rock Cycle

6.1 How Rocks Change

The rock cycle shows how rocks are continuously formed, changed, and reformed:

1. **Igneous to Sedimentary:** Weathering and erosion break down igneous rocks into sediments, which are then compacted and cemented to form sedimentary rocks.
2. **Sedimentary to Metamorphic:** Heat and pressure deep within the Earth change sedimentary rocks into metamorphic rocks.
3. **Metamorphic to Igneous:** Metamorphic rocks can melt into magma, which cools and solidifies to form igneous rocks.
4. **Any rock type can become any other rock type** through various processes in the cycle.

7 Fossils

7.1 What are Fossils?

- Preserved traces, imprints, or remains of ancient organisms
- Most commonly found in sedimentary rocks
- Provide evidence of past life and environments

7.2 Types of Fossils

- **Body Fossils:** Actual remains (bones, shells)
- **Trace Fossils:** Evidence of activity (footprints, burrows)
- **Imprint Fossils:** Impressions in rock

8 Common Misconceptions and Pitfalls

8.1 What is NOT a Mineral?

- **Living materials:** Ivory, amber, pearls
- **Man-made materials:** Plastic, glass, cement
- **Volcanic glass (obsidian):** Lacks crystalline structure
- **Rocks:** Mixtures of minerals (like sandstone)

8.2 Identification Mistakes to Avoid

- Relying only on color for identification
- Confusing similar-looking minerals (gold vs. pyrite)
- Not testing multiple properties
- Ignoring streak color vs. surface color

9 Study Tips and Assessment

9.1 Key Points to Remember

1. Always test multiple properties for mineral identification
2. Streak is more reliable than color
3. Hardness tests help distinguish similar minerals
4. Rocks are made of minerals, minerals are pure substances
5. The rock cycle is continuous and ongoing

9.2 Lab Activities

- Mineral identification stations with unknown samples
- Streak tests using streak plates
- Hardness tests using common objects
- Acid tests on calcite samples
- Rock cycle modeling with clay or playdough

9.3 Assessment Focus

- Understanding mineral definition (5 key characteristics)
- Proper use of identification tests
- Distinguishing between minerals and rocks
- Explaining the rock cycle
- Understanding weathering, erosion, and deposition